

OBSERVATIONS

- RMSProp optimizer converged faster than standard SGD and produced smoother loss reduction during training.
- Momentum-based optimization reduced oscillations compared to SGD and improved convergence stability.
- Cross-entropy loss performed better than Mean Squared Error (MSE) for the classification task, achieving higher validation accuracy and faster convergence.
- MSE loss learned more slowly because it is less suitable for multi-class classification problems compared to cross-entropy loss.
- A learning rate of 0.001 gave faster convergence, while 0.0001 produced slower but more stable training.
- Batch size 64 provided a good balance between training speed and stability.
- Validation accuracy improved steadily during initial epochs and started plateauing after a few epochs, indicating convergence of the model.
- ReLU activation helped the network train faster compared to sigmoid and tanh because it reduced vanishing gradient issues.
- Xavier/He initialization improved training stability compared to completely random initialization.
- Increasing hidden layer size improved representation learning up to a certain point, after which gains became marginal while computation cost increased.
- RMSProp and adaptive learning methods achieved better validation accuracy in fewer epochs compared to SGD.
- The confusion matrix showed that visually similar classes such as “Shirt”, “Pullover”, and “Coat” were more frequently misclassified.
- The model generalized reasonably well because training accuracy and validation accuracy remained close, indicating limited overfitting.
- Fashion-MNIST is more challenging than standard MNIST because clothing categories have higher visual similarity than handwritten digits.
- Hyperparameter tuning significantly affected performance, especially optimizer choice, learning rate, and hidden layer size.